

SHAPEIC: DESIGN-SPACE EXPLORATION

*Hierarchical LUT-based design-space exploration
for analog IC sizing*

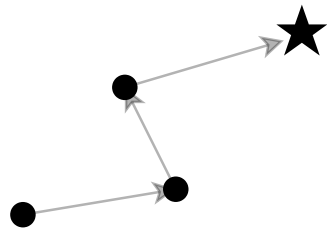
Daniel Arevalos

Munich University of Applied Sciences HM

Automatic Analog Sizing

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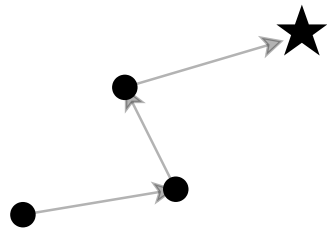
- ▶ Optimization-driven



- ✓ Strategic evaluation of candidates
- ✓ Many algorithms available

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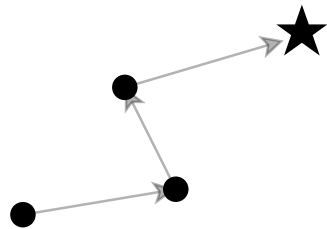
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- ✓ Strategic evaluation of candidates
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- ✗ Partial View of the design space
- ✗ May converge to local optima

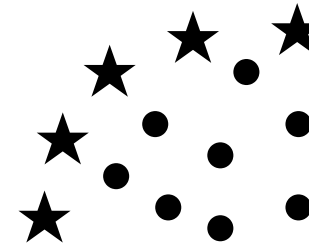
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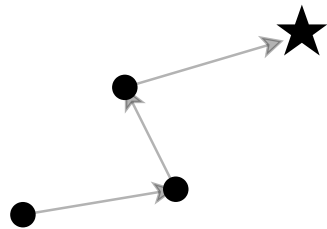
- ▶ Design-space exploration



- ✓ Design insight
- ✓ Alternative solutions

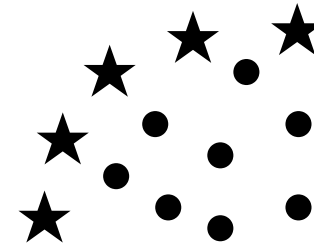
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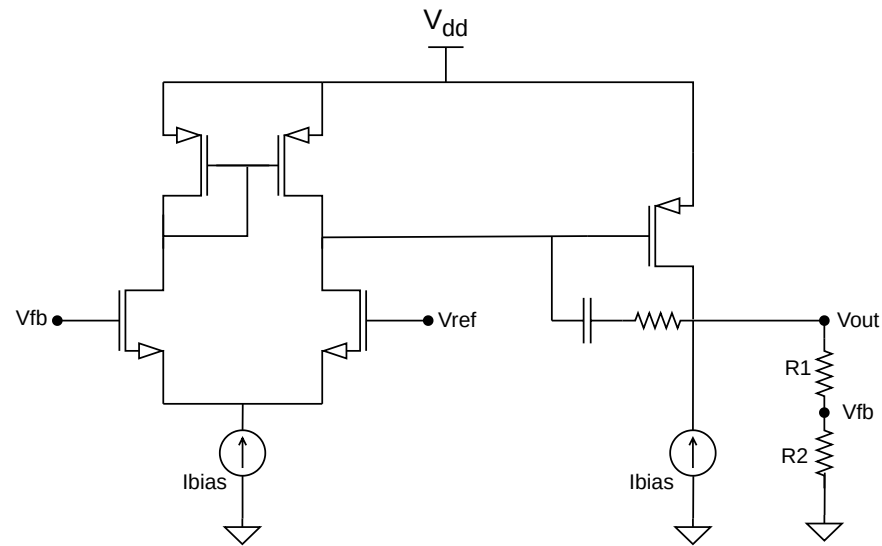
- ▶ Design-space exploration



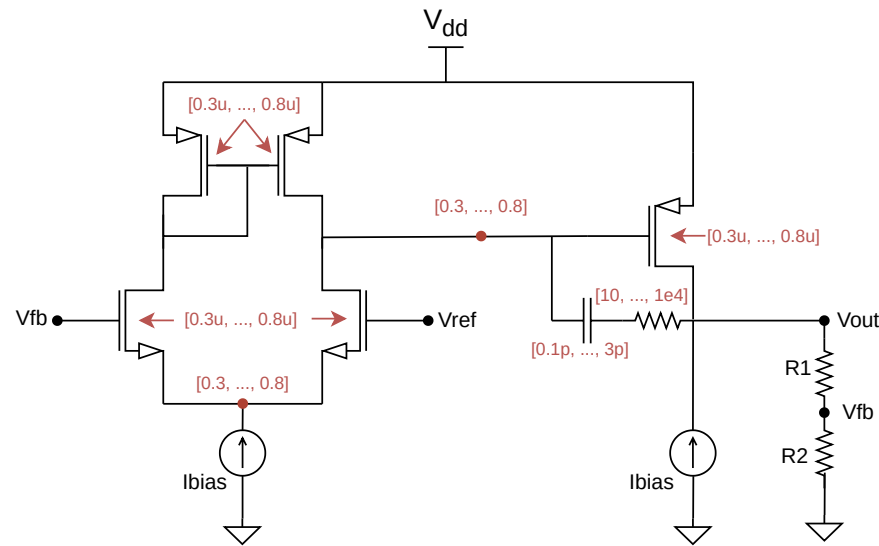
- ✓ Design insight
- ✓ Alternative solutions
- ✗ Expensive to build
- ✗ Scales poorly without abstraction

Let's explore a 2 Stage Amplifier

- ▶ Simple 5 transistor topology

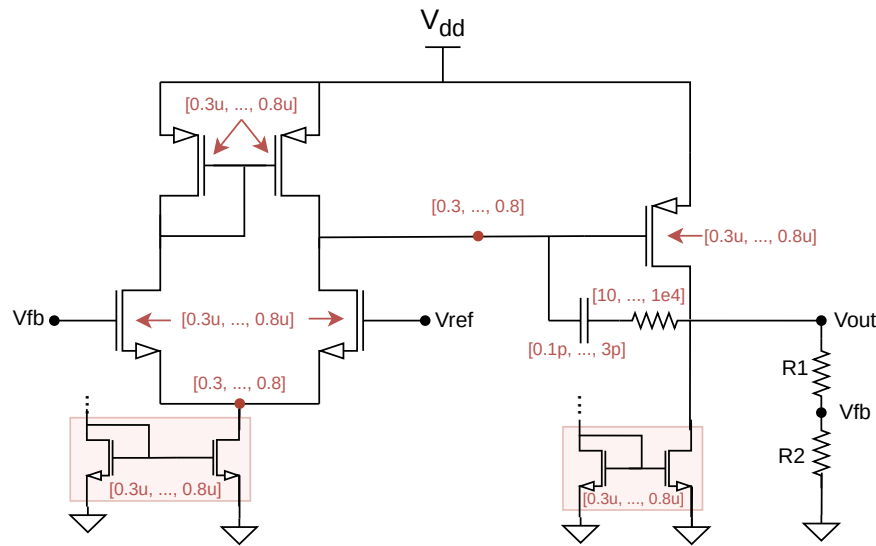


Let's explore a 2 Stage Amplifier



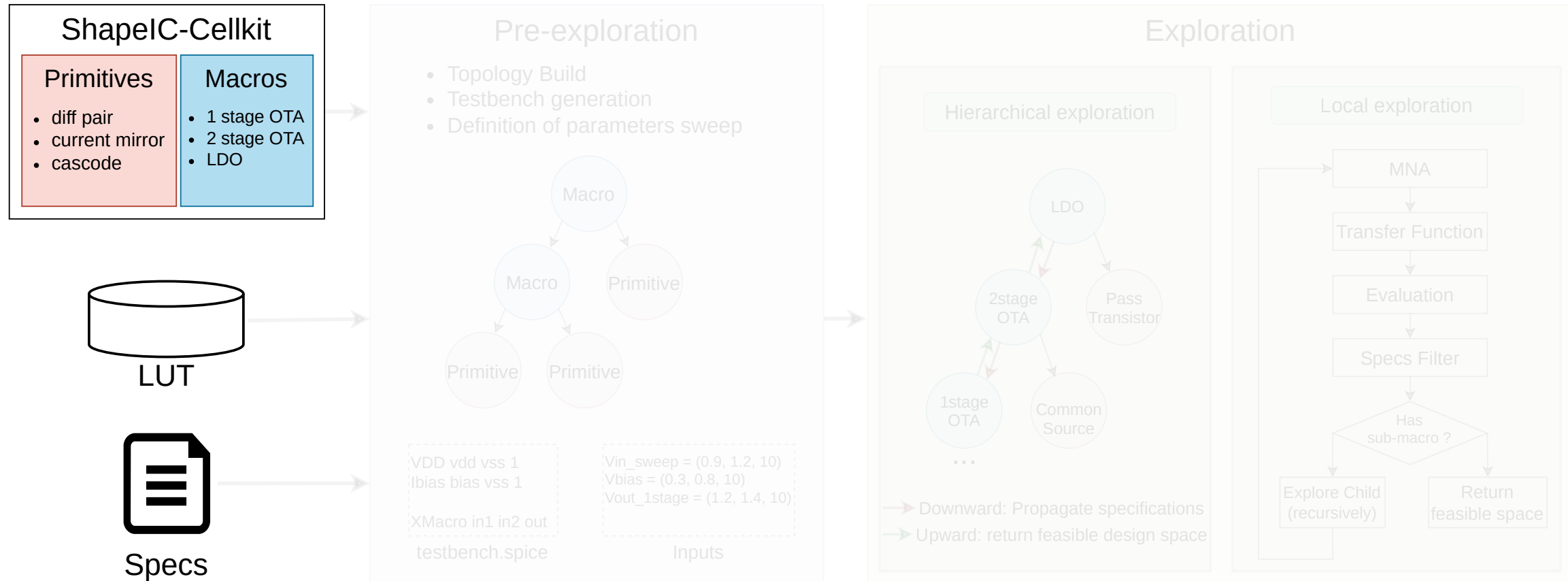
- ▶ Simple 5 transistor topology
- ▶ 1250000 design candidates
 - ▶ Simulation: $\approx 2h$
 - ▶ Equation evaluations: $\approx 2s$ (derived by hand)

Let's explore a 2 Stage Amplifier

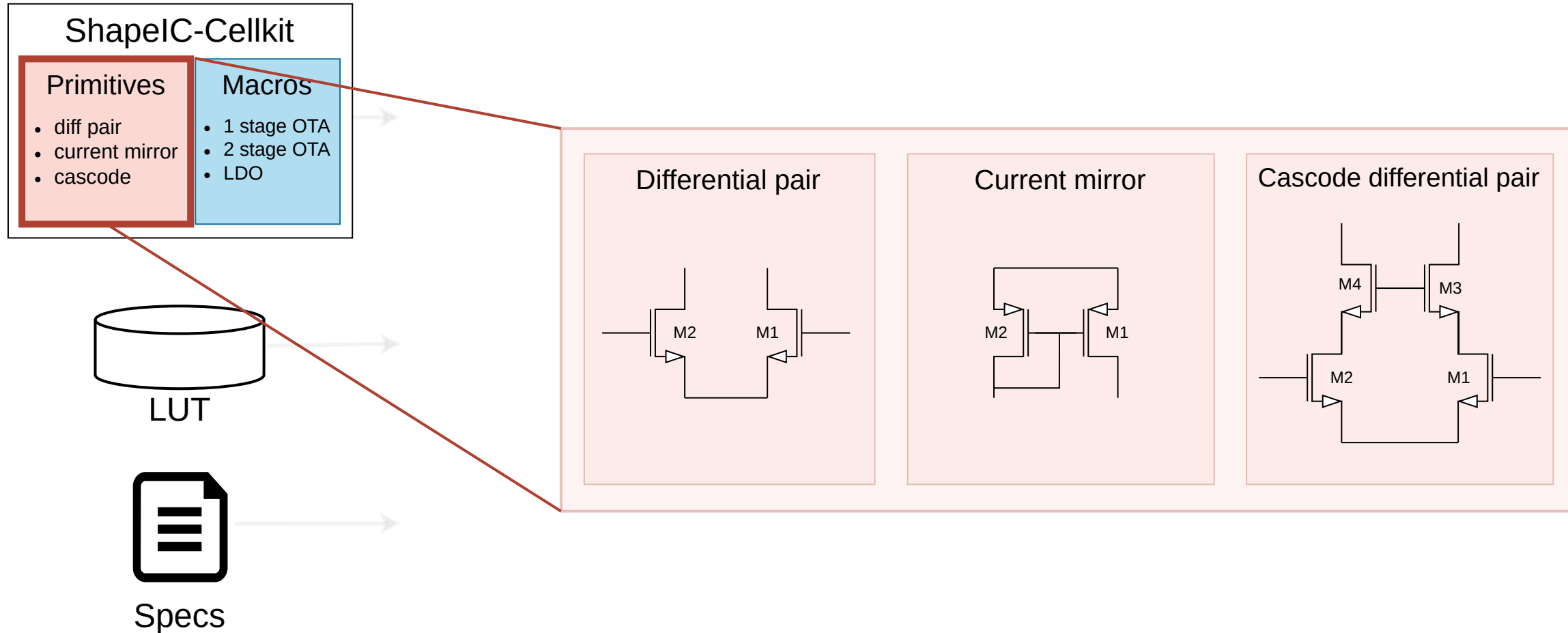


- ▶ Simple 5 transistor topology
- ▶ 1250000 design candidates
 - ▶ Simulation: $\approx 2h$
 - ▶ Equation evaluations: $\approx 2s$ (derived by hand)
- ▶ 31250000 design candidates
 - ▶ Simulation: $\approx 1month$
 - ▶ Equation evaluations: $\approx 1min$ (but you need to recalculate them)

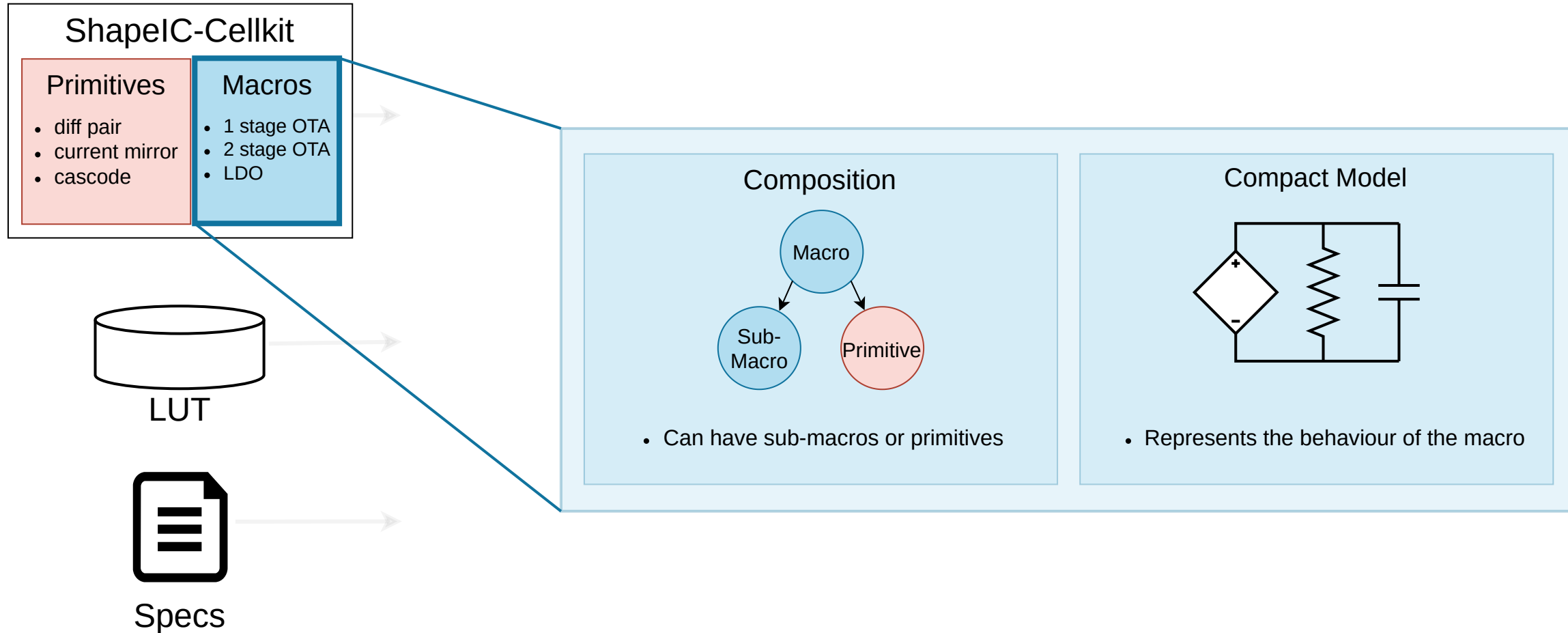
ShapelC: Building the Design Space



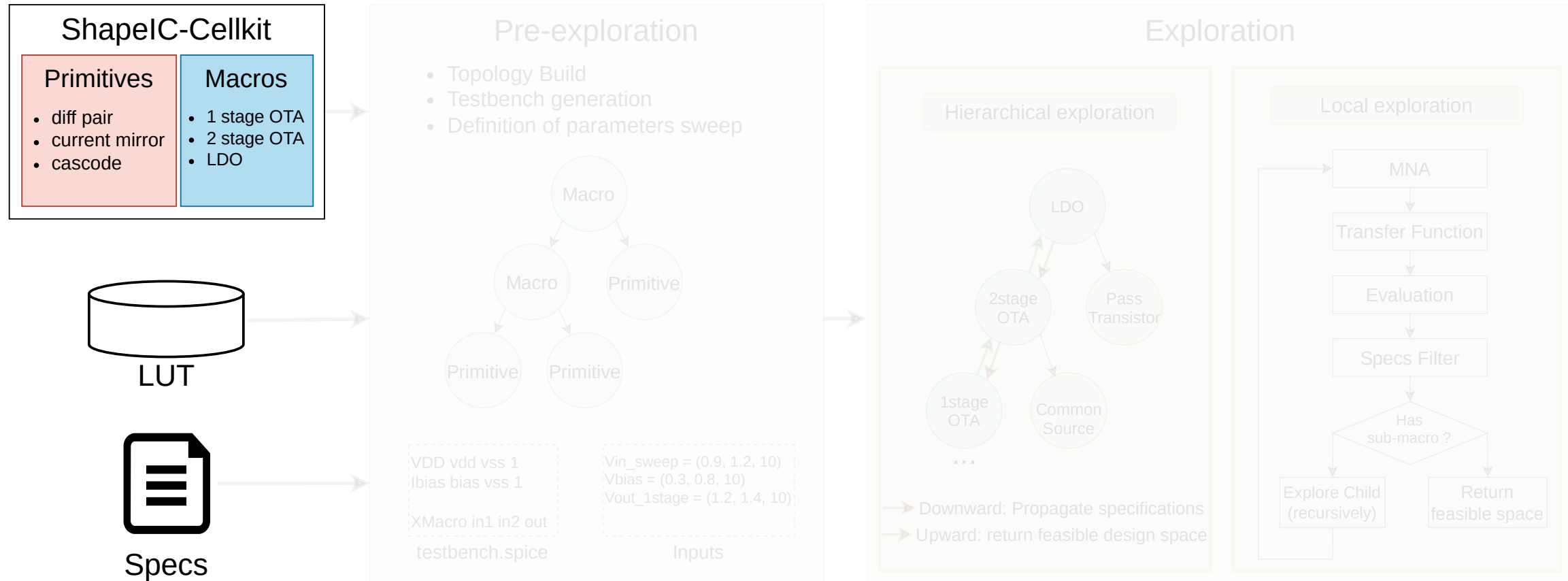
ShapelC: Building the Design Space



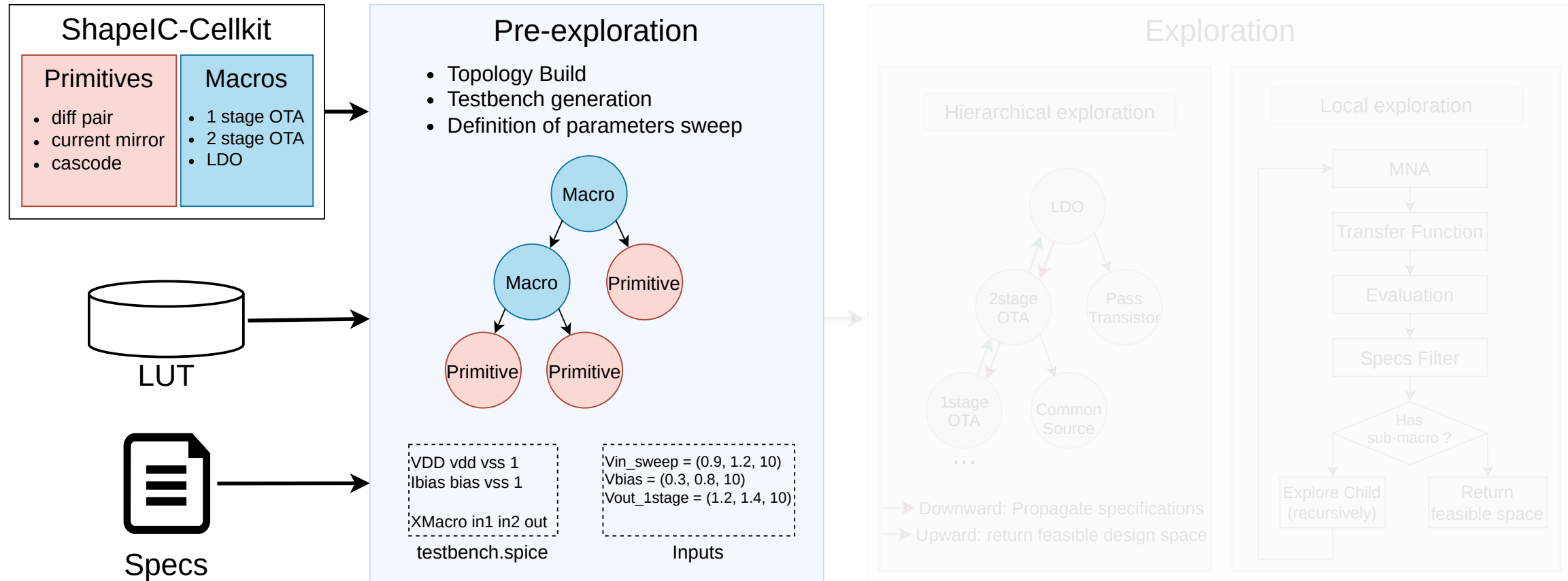
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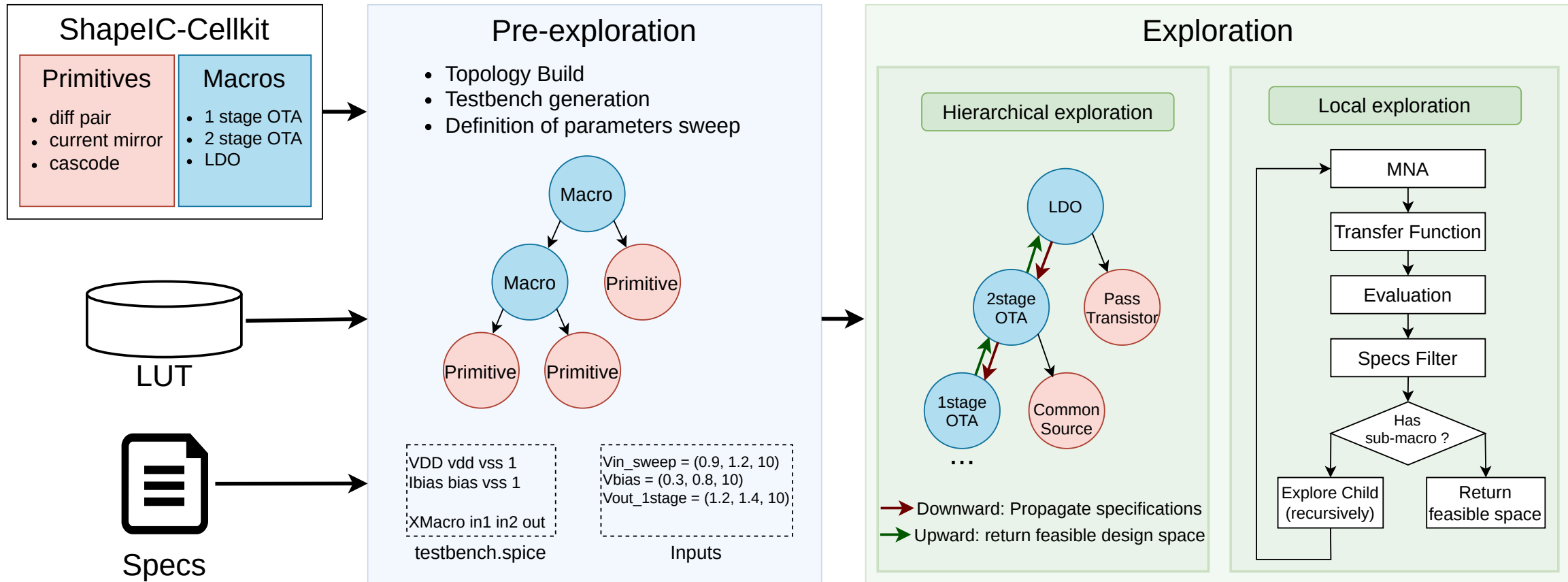
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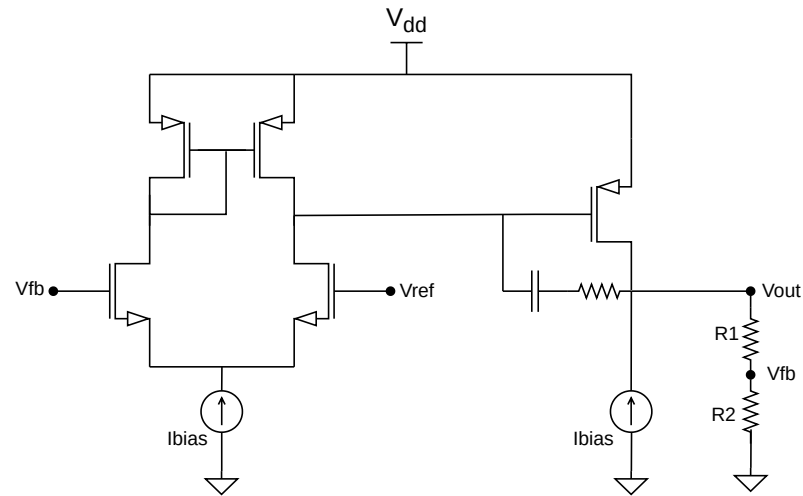
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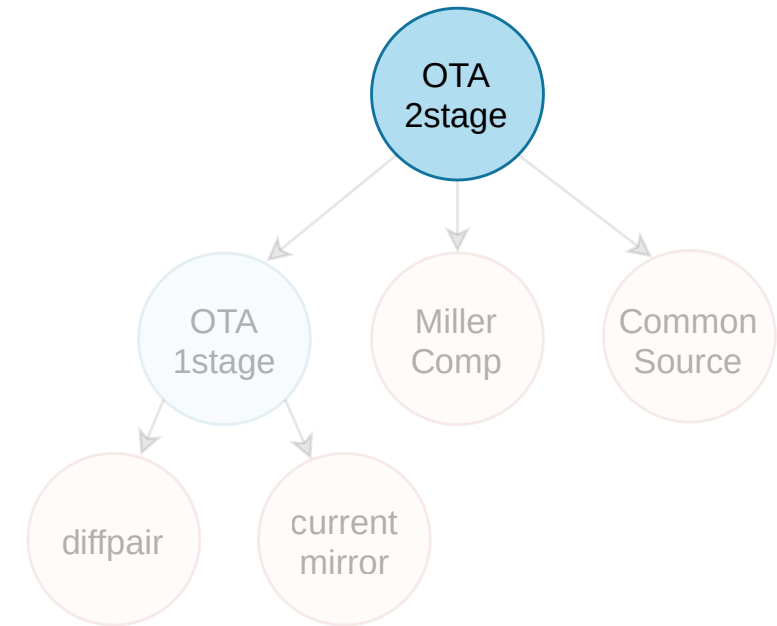
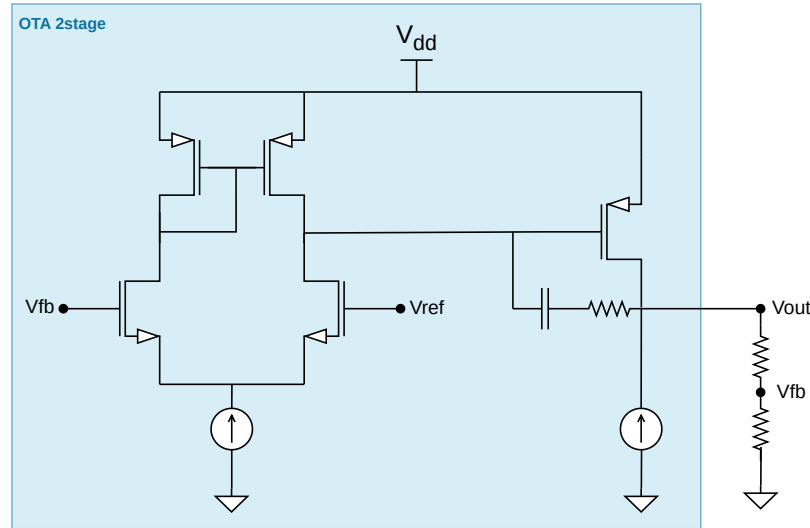
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ShapelC in Action: 2 Stage Amplifier (OTA)

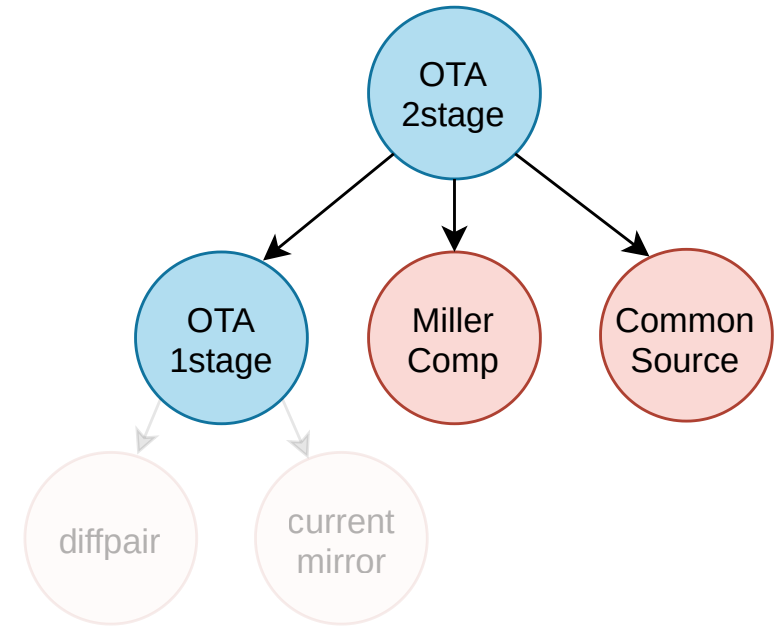
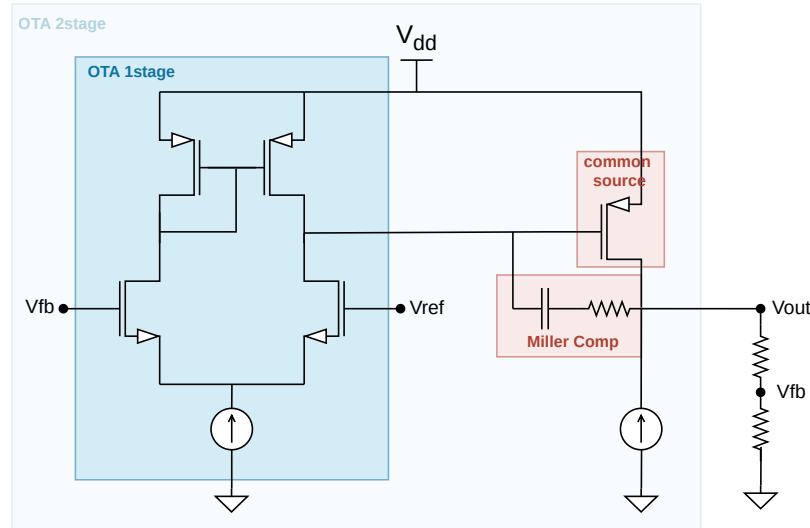


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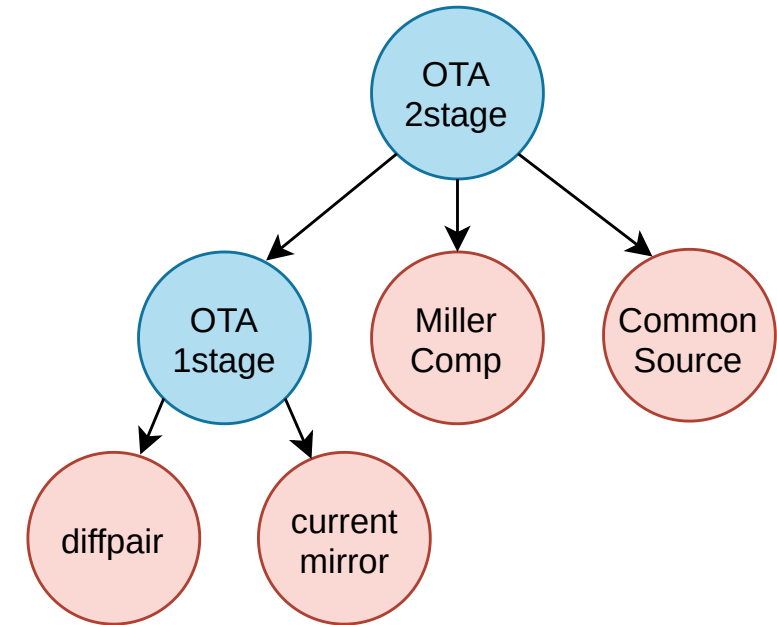
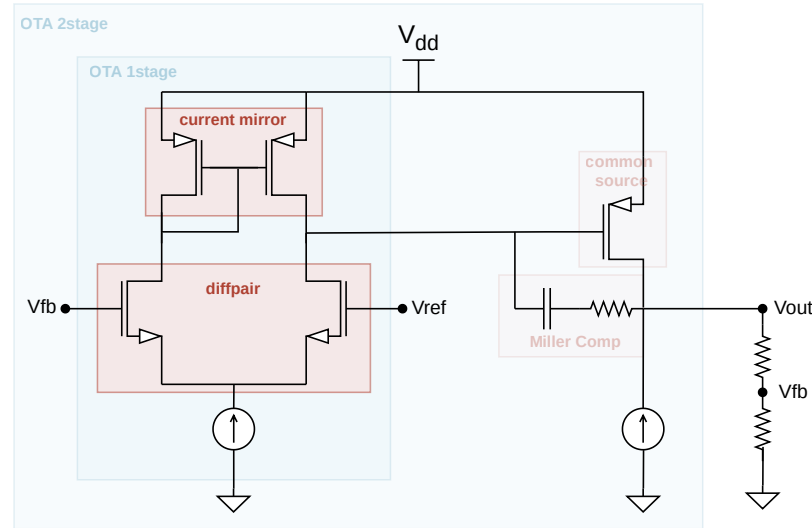
- ▶ Testbench: AC analysis
- ▶ Specs:
 $Gain \geq 70, BW \geq 10kHz, PM \geq 60$

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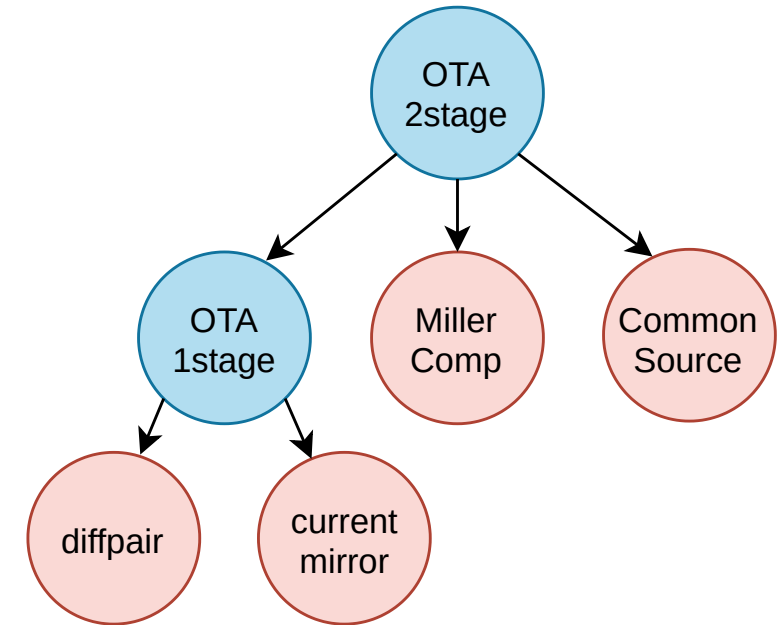
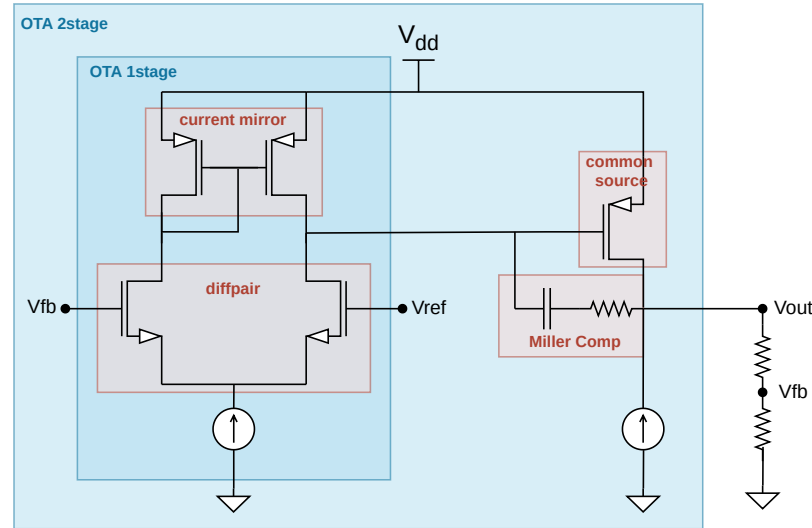
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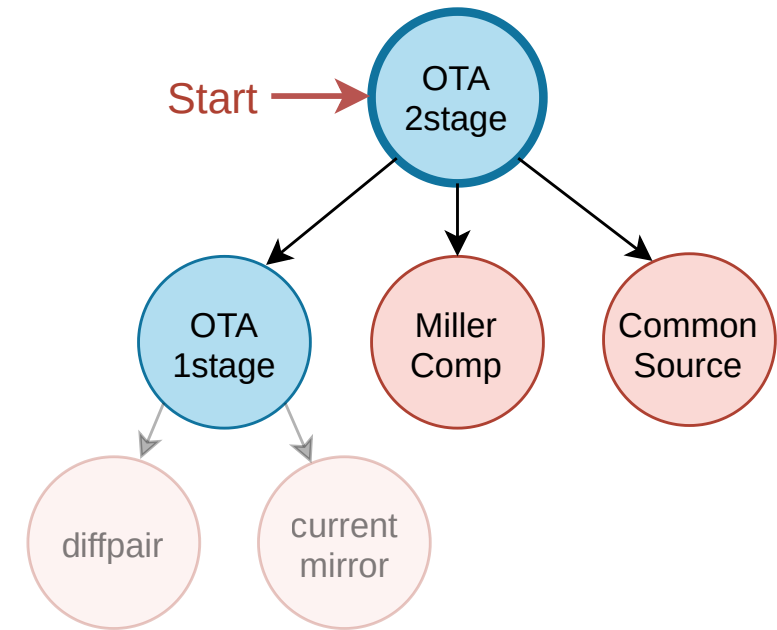


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1. Hierarchical exploration

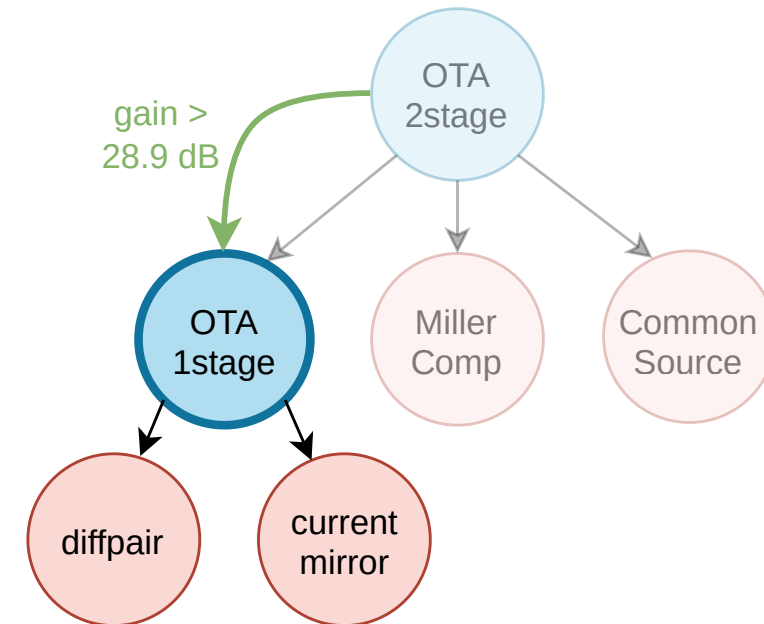
- ▶ TOP exploration: 500k → 2021
- ▶ Derived constraint: $Gain \geq 28.89 [dB]$



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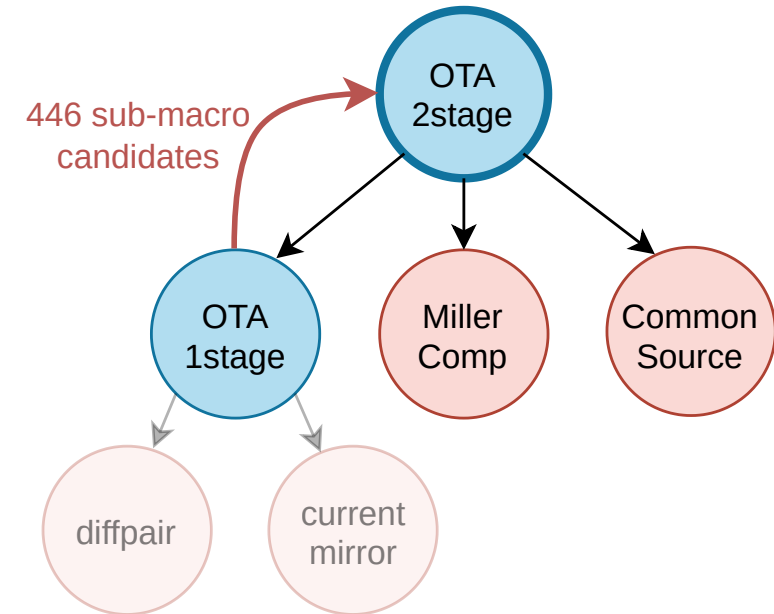
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- ▶ Sub-macro exploration: 1050 → 446
 - ▶ Return to parent 446 candidates



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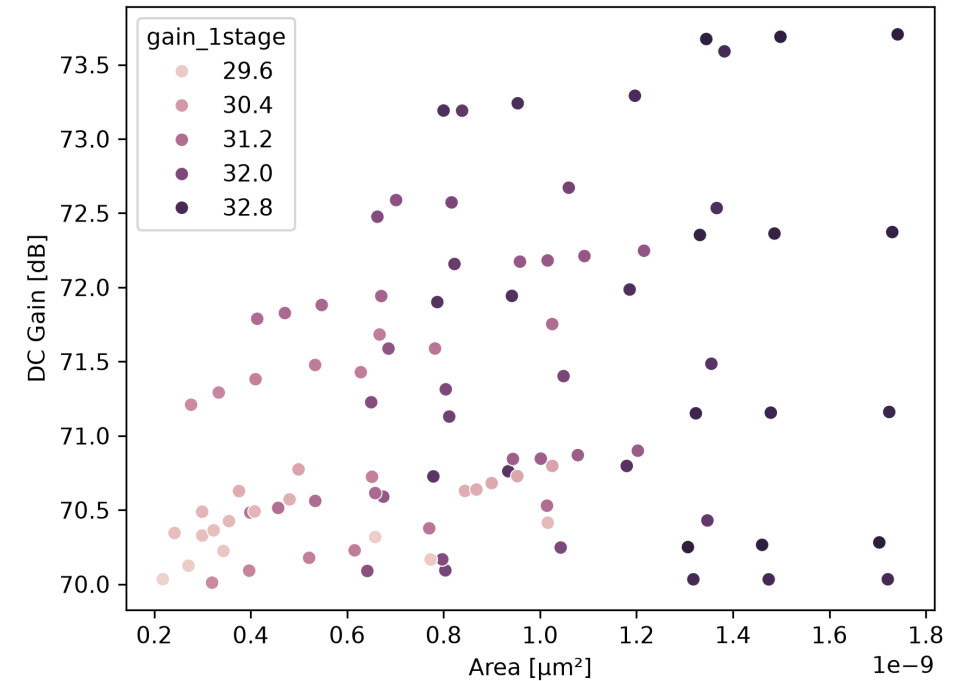
- ▶ TOP exploration: 500k → 2021
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- ▶ Sub-macro exploration: 1050 → 446
 - ▶ Return to parent 446 candidates
- ▶ TOP refresh: 223k → 224



ShapelC in Action: 2 Stage Amplifier (OTA)

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 - ▶ Return to parent 446 candidates
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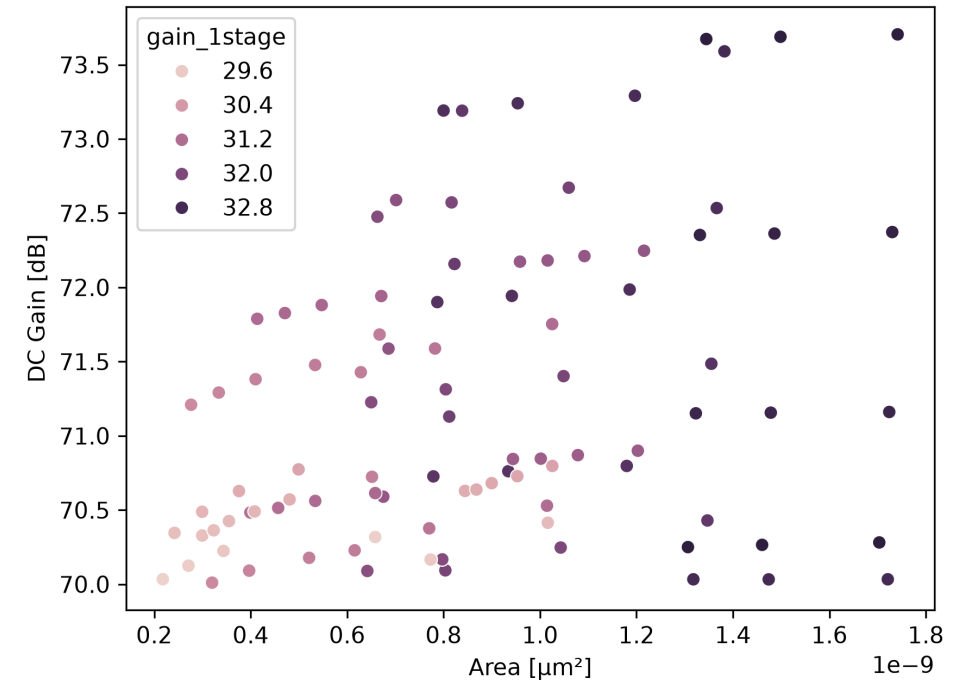
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2. Runtime

- ▶ Exploration only: $\approx 2.79s$
- ▶ Full run incl. LUT load: $\approx 8.38s$



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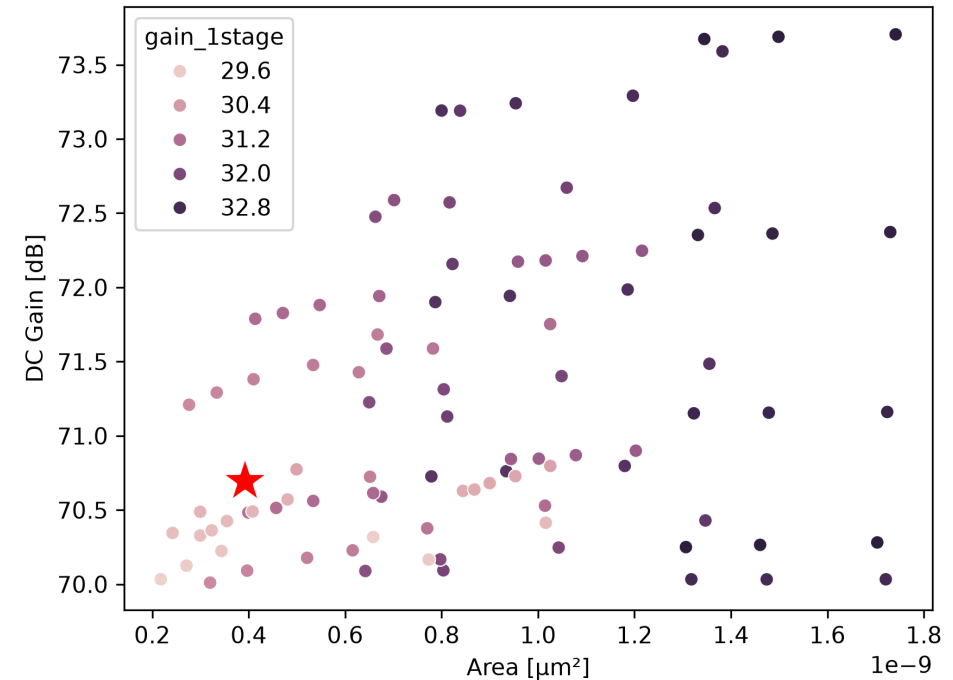
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- ▶ Exploration only: $\approx 2.79\text{s}$
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3. Simulation-based optimization

- ▶ Random initialization: no feasible solution found
- ▶ ShapelC-seeded initialization: converged to a feasible solution (★)



Current Status & What is Next ?

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- ▶ AC analysis supported
- ▶ Hierarchical LUT-based exploration
- ▶ Layout-aware exploration prototype
- ▶ ShapelC-Cellkit integration

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